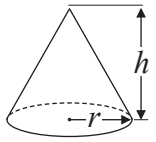
15 A cone has a volume of 98 cm^3.
The radius of the cone is 5.13 cm.

(a) Work out an estimate for the height of the cone.

Volume of cone = $\frac{1}{3}\pi r^2 h$



.....cm

(3)

John uses a calculator to work out the height of the cone to 2 decimal places.

(b) Will your estimate be more than John's answer or less than John's answer?
Give reasons for your answer.

.....

.....

.....

(1)

(Total for Question 15 is 4 marks)

16 n is an integer greater than 1

Prove algebraically that $n^2 - 2 - (n - 2)^2$ is always an even number.

×	n	−2
n	n^2	−2n
−2	−2n	+4

$(n - 2)^2 = n^2 - 2n - 2n + 4$

1 mark
Expand the bracket correctly

$n^2 - 2 - (n^2 - 4n + 4)$

$= n^2 - 2 - n^2 + 4n - 4$

1 mark
Subtract every term in the bracket

$= 4n - 6 = 2(2n - 3)$

1 mark
Collect and factor out 2

$2n - 3$ is an integer, so $2(2n - 3)$ is always even.

1 mark
A multiple of 2 for integer n

(Total for Question 16 is 4 marks)

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